

Technical note: Accuracy of an ear tag-attached accelerometer to monitor rumination and feeding behavior in feedlot cattle¹

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ABSTRACT: Early identification of sick cattle increases treatment success and decreases mortality. Continuous automated records of behavior can be used to identify sick cattle early in the disease process. The objective was to evaluate accuracy of an ear-attached accelerometer (SensOor) that quantified ear movements and estimated feeding and rumination time through a proprietary algorithm. Accelerometers were attached to the ear tag of 18 steers with an initial mean BW of 326 ± 46 kg. The manufacturer's proprietary software was used to determine time spent "feeding," "ruminating," "active," and "resting." Direct visual observation was used to validate the accelerometer. Sensitivity, specificity, and predictive values were calculated for rumination and feeding separately. Repeated measures were accounted for

using mixed model logistic regression. Single minutes of either feeding or rumination in a run of other behavior minutes were changed to the preceding behavior. Accuracy and precision of hourly recorded feeding and rumination times were assessed using the concordance correlation coefficient adjusted for repeated measurements. Sensitivity and specificity were 95 and 76% for feeding and 49 and 96% for rumination, respectively. Concordance correlation between observations and the sensor were 0.79 (95% CI: 0.61 to 0.85) and 0.44 (95% CI: 0.23 to 0.60) for feeding and rumination, respectively. There was large variability among steers, with concordance correlations ranging from 0.09 to 0.98 for rumination time and from 0.58 to 0.96 for feeding time. We conclude that the accelerometer is a promising monitoring system for feeding behavior.

Key words: automation, behavior monitoring, cattle, feeding, rumination, validation

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INTRODUCTION

Feeding and rumination behavior can be monitored to assess health status of ruminants. For example, records of feeding behavior have been used to detect bovine respiratory disease in feedlot cattle (Sowell et al., 1999; Borderas et al., 2008), and monitoring rumination behavior helped detect subacute ruminal acidosis in dairy cows (DeVries et al., 2009). However, behavioral assessments using visual appraisal are very time consuming and typically done only once or

twice daily for relatively short intervals. Consequently, monitoring devices have been developed to continuously measure either feeding (Mendes et al., 2011) or rumination behavior (Elischer et al., 2013). Such devices are based on radio-frequency identification (**RFID**) at the feed bunk (Schwartzkopf-Genswein et al., 2011), movements of the jaw (Beauchemin et al., 1989), or acoustics (Schirrmann et al., 2009).

Based on observed differences in the movement patterns of cows' ears, a system has been developed to monitor ear movements and calculate feeding, rumination, activity, and resting times (SensOor; Agis Automatisering BV, Harmelen, The Netherlands). A 3-dimensional accelerometer is attached to the RFID ear tag, and an online application provided by the manufacturer records time spent feeding, ruminating, active, and resting per hour and per day. A validation study in dairy cows suggests that this accelerometer was accurate for calculating feeding and rumination times (Bikker et al.,

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2014). However, some monitoring systems, which have high accuracy in dairy cattle, do not perform as well in beef cattle (Goldhawk et al., 2013). The objectives of this study were to evaluate 1) the sensitivity and specificity of the accelerometer to detect rumination and feeding behavior in beef cattle and 2) the accuracy of time spent ruminating and feeding calculated by the accelerometer.

MATERIAL AND METHODS

The University of Calgary Animal Care Committee reviewed and approved all procedures used (AC13-0104). The study was conducted in August 2013 at the Agriculture and Agri-Food Canada, Lacombe Research Centre (Lacombe, AB, Canada). Eighteen Hereford $\times$ Angus yearling steers with an initial mean BW of 326 kg (SD $\pm$ 46 kg) were group-housed in an outdoor dirt floor pen. Approximately 1/3 of the pen was sheltered and closed on 3 sides. Steers were slick-bunk fed with 100% barley silage once daily at 0830 h.

Two observers were trained to identify feeding and rumination. The observers discussed and agreed on the definition of rumination and feeding prior to the start of the experiment. First, observers watched video recordings of rumination and feeding. Second, both observers monitored and discussed the behavior of the same 2 steers in the pen for 2 h. The onset of a rumination bout was defined as the time when regurgitation occurred, namely when a bolus came up the esophagus and reached the mouth (Schirmann et al., 2009). The end of a rumination bout was the minute the last bolus was swallowed (>1 min between 2 boluses was considered a new rumination bout). Observers recorded the start time for feeding as soon as the steer had its head over the feed bunk and started licking and chewing movements. The end of the feeding bout was defined when the steer stepped back from the feed bunk and stopped licking and chewing (Bikker et al., 2014). If the head of the observed steer was not fully visible, the minutes of non-observation were marked and excluded from analysis ($n = 82$ min). Observers recorded behavior 2 or 3 days per week during the study period (total, 13 d). Observation periods lasted between 72 and 240 min (mean 126 min). The earliest observation started at 0430 h, and the latest finished at 2200 h. Every observation period, 2 steers were selected randomly for observation, with no steer being observed >9 times. On average, each steer was observed for 5 (SD $\pm$ 2) observation periods (Table 1).

Interobserver variability was measured by having both observers independently record the behavior of the same 2 steers for 2 h, 3 times during the study period (total of 6 h). The observers did not communicate and could not view each other's records. Pearson's correlation coefficient for paired observations of feed-

Table 1. Observational times in minutes (percentage of total) per steer

Steer no.	Feeding	Ruminating	Total time
1	243 (25)	270 (28)	965
2	109 (31)	48 (14)	355
3	91 (22)	83 (20)	409
4	100 (42)	88 (37)	238
5	223 (36)	79 (13)	625
6	117 (24)	151 (32)	478
7	153 (26)	132 (23)	578
8	119 (14)	117 (14)	858
9	93 (15)	136 (21)	640
10	202 (30)	174 (26)	679
11	29 (24)	0 (0)	120
12	64 (16)	112 (27)	409
13	121 (28)	180 (41)	438
14	106 (20)	190 (37)	519
15	204 (33)	57 (9)	626
16	410 (42)	223 (23)	969
17	193 (22)	209 (25)	867
18	115 (20)	138 (25)	561

ing bouts ($n = 58$) and rumination bouts ($n = 16$) between the 2 observers to assess interobserver reliability was $r > 0.98$ for all comparisons.

Feeding and rumination times were monitored continuously using the accelerometer, which was mounted to the RFID tag (Allflex, Dallas, TX) and placed in the proximal half of the ear between the 2 cartilage folds. Raw data were transmitted every minute through radio frequency technology (ZigBee Alliance, San Ramon, CA) via router and coordinator to a computer in the office. The working frequencies of RFID tags and accelerometers (134.2 kHz and 2.4 GHz, respectively) did not interfere. An online application (Cowmanager; Agis Automatisering BV) classified each minute with a proprietary algorithm as 1 of 4 behavior categories: "feeding," "ruminating," "active," and "resting."

Live observations were used as a reference standard for comparison with the accelerometer. Statistical analyses and data editing were performed using Stata Version 13.1 (StataCorp LP, College Station, TX). Single feeding or rumination minutes in a run of other behavior were converted to the preceding behavior by applying a filter in the accelerometer data (Ledgerwood et al., 2010).

The accuracy of the accelerometer was tested qualitatively and quantitatively. First, sensitivity, specificity, and positive and negative predictive values were calculated to assess the accuracy of feeding and ruminating per minute. Steer identification was included as a random effect in a mixed logistic regression to account for clustering within individuals (Genders et al., 2012). Second, records were summarized hourly to assess ac-

Table 2. Comparison between observed (row) feeding, rumination, and other behavior (others) and unadjusted accelerometer-recorded (SensOor; Agis Automatisering, Harmelen, The Netherlands) feeding, ruminating, active, and resting behavior (column) per minute (percentage of observed behavior how it was recorded by accelerometer)

Observation	Accelerometer				Total
	Feeding	Ruminating	Active ¹	Resting	
Feeding	2,488 (93)	61 (2)	139 (5)	4 (0)	2,692 (100)
Ruminating	768 (32)	1,212 (51)	352 (15)	55 (2)	2,387 (100)
Others	1,400 (27)	549 (11)	2,236 (43)	988 (19)	5,173 (100)
Total	4,656	1,822	2,727	1,047	10,252

¹Head movements besides feeding or ruminating.

curacy of quantitative outcomes (duration of feeding and rumination). The R package (R Foundation for Statistical Computing, Vienna, Austria) CCCRM was used to calculate concordance correlation coefficient (CCC) on hourly-summarized records adjusted for random steer effect to compare results of visual versus accelerometer-recorded data (Carrasco et al., 2013). In comparison to simple Pearson's correlation, concordance correlation reflects the level of agreement including a location-shift parameter and a scale-shift parameter that measures the difference between the slope and perfect correlation (45° angle). Bland-Altman graphs were used for graphical representations of the residuals for paired measurements (Dohoo et al., 2009).

RESULTS

Observers recorded 10,252 min of behavior on 18 steers distributed over 13 d during daylight hours (Table 2). Of the total observation time, steers were observed as feeding for 2,692 min (26%) and ruminating for 2,359 min (23%). Mean observed feeding time across all steers was 10.8 ± 15.0 min/h, and average rumination time was 9.5 ± 13.9 min/h. When a 1-min filter was applied, sensitivity increased from 93

to 95% (95% CI: 93 to 96%) and 48 to 49% (95% CI: 34 to 64%) for feeding and rumination, respectively (Table 3). Furthermore, specificity increased from 70 to 76% (95% CI: 69 to 81%) and 94 to 96% (95% CI: 94 to 98%).

Concordance correlations between observed and accelerometer-recorded feeding and rumination time were CCC = 0.79 and 0.44. On average, observed feeding was 6.6 min (± 8.7 min) shorter and rumination 2.7 min (± 9.7 min) longer per observed hour than estimated by the accelerometer (Fig. 1). Specifically, when no filter was applied, the accelerometer misclassified 32% of observed rumination minutes as feeding minutes, 15% as "active" and 2% as "resting." Conversely, 2% of observed feeding minutes were classified as rumination by the sensor (Table 2). Variability among steers ranged from concordance correlations 0.09 to 0.98 for rumination time and from 0.58 to 0.96 for feeding time.

DISCUSSION

The ear tag-attached accelerometer provided good sensitivity but only moderate specificity to estimate feeding. On the contrary, rumination monitoring by the accelerometer had high specificity but low to moderate sensitivity. Therefore, the accelerometer seemed to be a promising tool to measure feeding behavior in beef cattle but needs improvement for measuring rumination.

The good concordance correlation between observed and calculated feeding time by the accelerometer was in agreement with a concordance correlation of 0.75 between observed and calculated feeding time in dairy cattle assessed with the same technology (Bikker et al., 2014). Other technologies, including the GrowSafe system (Growsafe Systems Ltd., Airdrie, AB, Canada) and the Insentec monitoring system (Insentec, Marknesse, The Netherlands), have been validated for recording feeding behavior in beef cattle. Although these systems are highly accurate and have the ability to measure intake (Chapinal et al., 2007; Schwartzkopf-Genswein et al., 2011), they are based

Table 3. Test characteristics (95% CI) of an ear tag-attached accelerometer (SensOor; Agis Automatisering BV, Harmelen, The Netherlands) to monitor feeding and rumination in feedlot cattle adjusted for clustering within steer

Behavior (10,252 min)	Sensitivity	Specificity	PPV ¹	NPV ²	CCC ³
No filter					
Feeding (2,690 min)	93 (91–94)	70 (65–75)	54 (48–59)	97 (95–97)	0.75 (0.61–0.84)
Rumination (2,462 min)	48 (35–63)	94 (91–95)	66 (57–74)	87 (84–91)	0.41 (0.20–0.58)
One-minute filter					
Feeding (2,690 min)	95 (93–96)	76 (69–81)	60 (54–65)	98 (97–98)	0.79 (0.61–0.85)
Rumination (2,462 min)	49 (34–64)	96 (94–98)	78 (69–85)	88 (82–91)	0.44 (0.23–0.60)

¹Positive predictive value.

²Negative predictive value.

³Concordance correlation coefficient.

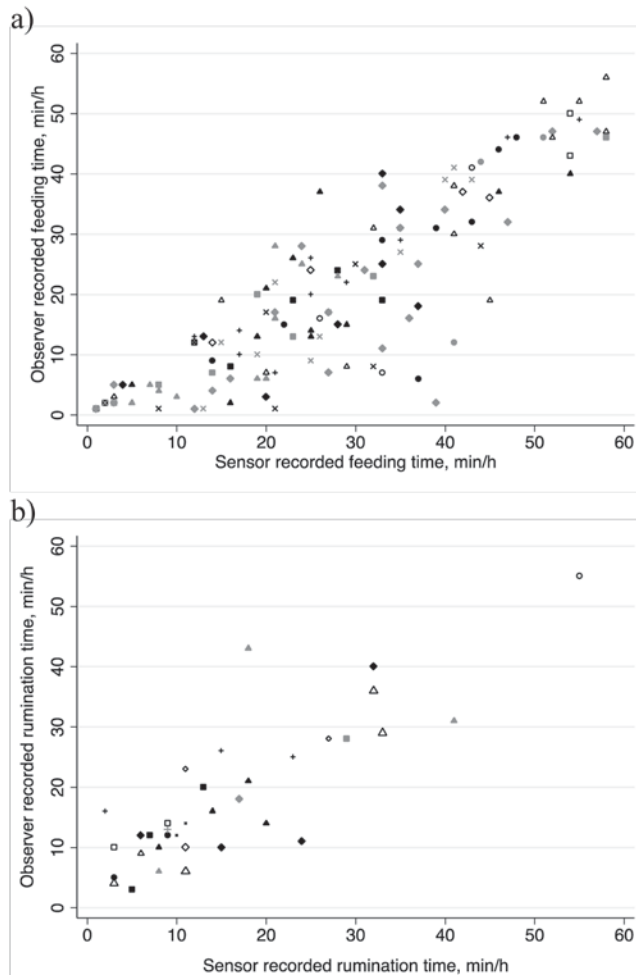


Figure 1. Number of minutes that individual steers spent a) feeding and b) ruminating during 1 h recorded by observers (y) and the accelerometer (x). Note that individual steers are represented in various shapes and colors.

on individual feed bunks, allowing only single-animal access to a limited number of bunks. Conversely, the accelerometer enabled monitoring of natural cattle behavior, feeding simultaneously (Rook and Huckle, 1995), and does not require any adaptation period. Notwithstanding, an important limitation is that feed intake cannot be measured with the accelerometer.

Concordance correlation for rumination monitoring compared to observations was low to moderate. Conversely, in dairy cattle, the same system had a high concordance correlation ($CCC = 0.93$) between observed rumination times and estimates by the accelerometer (Bikker et al., 2014). However, the cattle used in the current study were fed a different diet (barley silage vs. partially mixed ration or total mixed ration), differed in breed (Hereford-Angus vs. Holstein Friesian), sex (male castrated vs. female), and age (yearling vs. lactating dairy cow) compared to the previous study, which could have accounted for the differences. Similar to this study, an acoustic sensor to monitor rumination was tested in dairy cattle and had a high correlation to visual observation (r ranged from

0.86 to 0.96; Schirmann et al., 2009). When tested on beef cattle (Goldhawk et al., 2013) and younger Holstein cattle (<9 mo; Burfeind et al., 2011), the correlation was only $0.24 (\pm 0.07)$ and between 0.47 and 0.89, which changed with age, respectively. This previously reported lack of correlation, therefore, highlights the differences between the 2 types of cattle.

The interanimal variation in correlation between estimates provided by observations and the accelerometer can only speculatively be attributed to ear movement differences between cattle. Whereas most of the steers had concordance correlations between 0.41 and 0.98, the concordance correlation in 1 individual was as low as 0.09 for rumination. When age, weight, and individual standardized head pictures of the steers taken in the chute at processing were compared, no explanation for the variation could be suggested. Interestingly, Burfeind et al. (2011) reported similar variation of accuracy among young calves when they used acoustic methods to monitor rumination. However, although the accelerometer did not provide perfect estimates in all steers, it predicted rumination better than a previously validated system in most beef cattle (Goldhawk et al., 2013). Furthermore, it is noteworthy that there are currently few commercial options to monitor rumination.

The moderate specificity of feeding monitoring in the present study might be explained by head movements associated with fly defense, as cattle use their heads, ears, tails, legs, and skin movements to defend themselves from flies (Dougherty et al., 1993). Ruminating steers frequently laid down in the shaded area with limited ventilation compared to the feed bunk, which was situated in the open part of the pen. This resulted in a higher fly burden during rumination compared to feeding. Additionally, a high proportion of rumination minutes were classified as feeding by the accelerometer. Based on the high interobserver correlation, however, we concluded that errors due to observers' misinterpretations were minimal.

This study focused on evaluating the sensitivity, specificity, and accuracy of the accelerometer to measure feeding and rumination behavior, as most applicable to the beef industry. However, since the system additionally records activity and resting, these should be evaluated in future studies. Further, for application in feedlot cattle at the end of the feeding period, evaluation of the accuracy in older feedlot cattle would be recommended.

In conclusion the ear-attached accelerometer is a promising tool to measure feeding behavior in beef cattle. Its associated algorithm, however, might need to be optimized to better differentiate rumination from feeding.

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